William McLean

AI Developer Advocate & Chief Technology Officer | Google Developer Expert (GDE) AI/Cloud | AWS Community Builder

Email: xbill@glitnir.com | Lawrenceville, NJ 08648

LinkedIn: linkedin.com/in/xbill | GitHub: github.com/xbill9 | Hugging Face: huggingface.co/xbill9 | Blog: dev.to/xbill | medium.com/@xbill999

Professional Summary

Developer Relations leader and CTO who has been **building software agents for 30+ years** — from IBM's agent-oriented OPC Tracker Agent (1992) and a master's thesis on extensible agent technology to production agentic AI systems built on Gemini, Vertex AI, the Model Context Protocol (MCP), and the A2A protocol. **Google Cloud Ambassador, Google Developer Expert (GDE) for AI and Cloud, AWS Community Builder (AI Engineer category), and Google TPU Builders participant**, holding 10 Google Cloud certifications. Published the **first community port of Gemma 4 to AWS Inferentia2 hardware** (Hugging Face). Author of **300+ technical articles** and 170+ open-source repositories spanning multi-agent systems, LLM serving on Cloud TPU and GPUs, and agentic coding tools including Claude Code, OpenAI Codex, and Gemini CLI.

Core Competencies & Skills

- **Developer Relations & Advocacy:** Developer Tools & SDKs, Community Engagement, Technical Content Creation (Blogs, Code Samples, Demos), Developer Programs, API Design & Review, Developer Feedback Loops, Speaking & Evangelism.
- **Artificial Intelligence (AI) & Machine Learning (ML):** Generative AI (GenAI) — Gemini, Gemma, Anthropic Claude, OpenAI GPT; AI Agents & Agentic Workflows, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vertex AI, Hugging Face, Model Context Protocol (MCP), Agent-to-Agent (A2A) Protocol, Agent Development Kit (ADK), Large Language Model (LLM) Orchestration, LLM Serving & Inference (vLLM, Cloud Tensor Processing Units / TPU v6e, AWS Inferentia2, NVIDIA L4), Model Quantization (QAT), Model Evaluation & Benchmarking, MLOps, TensorFlow, PyTorch, Automated Code Analysis.
- **Agentic Coding & AI Developer Tools:** Anthropic Claude Code (MCP integrations, skills, sub-agents), OpenAI Codex, Gemini CLI, Antigravity CLI, Kiro — building, porting, and benchmarking MCP servers and agent toolchains across all major AI coding platforms.
- **Cloud Architecture:** Google Cloud Platform (GCP Expert), AWS (Community Builder), Azure; Multi-Cloud Strategy, Serverless (Cloud Run, Lambda), Kubernetes (GKE, EKS), Terraform, CI/CD, Infrastructure as Code (IaC).
- **Technical Leadership:** CTO-level Strategy, Board Advising, Investor Relations, Cross-Functional Team Leadership, Strategic Planning, Influence without Authority.
- **Software Engineering:** Distributed Systems, API Design (REST/gRPC), Event-Driven Architecture, Microservices, High-Availability Systems, Client Libraries.
- **Languages:** Python, Rust, Go, TypeScript, Node.js, Java, PHP, C/C++, Bash, REXX.
- **Security & Compliance:** Zero-Trust Architecture, PCI-DSS Compliance (15+ years), PKI, OAuth, Network Security.

Certifications & Developer Programs

- **Google Cloud Ambassador (2026):** Selected for the Google Cloud Ambassadors program (North America) — recognized by Google Cloud for partner-ecosystem technical leadership and community advocacy.
- **Google Developer Expert (GDE):** AI and Cloud (current) — Representing the voice of developer communities, collaborating with Google product managers, and presenting live demos/keynotes globally.
- **TPU Builders (Google):** Deploy and benchmark Gemma-family models on Cloud TPU v6e as part of Google's TPU Builders program; publish TPU deployment guides and open-source MCP tooling for the community.
- **AWS Community Builder (2026):** Selected for the AI Engineer Category; five-time AWS Community Builder Spotlight honoree (5-Timers Club).
- **dev.to Top Azure Author:** Awarded the dev.to Azure badge, given to the top Azure author each week (dev.to/xbill).
- **Google Cloud Certified (10x):** Professional Agentic Architect (beta first cohort, certificate series #166), Professional Machine Learning Engineer, Generative AI Leader, Cloud Architect, Data Engineer, DevOps Engineer, Cloud Developer, Cloud Security Engineer, Associate Cloud Engineer, Digital Leader.
- **Google Cloud Partner Certifications (2026):** Gemini Enterprise Deployment (secure environment configuration, Google Workspace data integration, Model Armor, search governance) and Gemini Enterprise Agent Development (AI agent development, API integration, prompt tuning, Antigravity IDE).

Professional Experience

Developer Advocate | Google Developer Expert (AI/Cloud)

Google Developer Program, New York, NY

2024 - Present

- Represent developer communities to Google product teams through direct feedback channels with Google Cloud and Google DeepMind engineering — reviewing emerging API designs, testing pre-release features, and advocating for developer interests.
- Deploy and benchmark Gemma-family models on Cloud TPU v6e and GPU infrastructure as a **Google TPU Builders** participant, publishing deployment guides and open-source MCP tooling for the community.
- Delivered **10+ keynotes, conference talks, and workshops** in the past year on agentic AI, multimodal agents, and LLM deployment (Build with AI, DevFest, university GDG chapters, hackathons).
- Facilitated hands-on developer workshops as a technical assistant (TA), including a post-event workshop on the **Google Cloud Next 2026** "Agentic Cloud" keynote demos and companion codelabs; sample **currency-agent** (A2A + ADK + MCP) featured in a Google codelab.

Co-Founder & Chief Technology Officer

Glitnir Ticketing, New York, NY

2002 - Present

- **Developer Enablement & Tooling:** Authored the open-source **"ADK Agent Starter Kit"** and developed high-impact AI utilities, including **mcp-on-cloudrun** (high-performance MCP deployments), to simplify the creation of agentic workflows and client-side implementations for thousands of developers.
- **AI Transformation:** Architected an AI-first engineering culture by integrating agentic coding tools — Claude Code, OpenAI Codex, Gemini CLI — and MCP across the software development lifecycle, from code review and testing to legacy migration.
- **Cloud Infrastructure:** Designed and managed a multi-cloud (GCP & AWS) enterprise SaaS platform processing **$20M+ in annual transactions** with **99.9% availability**.
- **Regulatory Compliance:** Engineered and maintained a zero-trust architecture, securing **15+ consecutive years of PCI-DSS compliance**.
- **Legacy Modernization:** Developed **refactor**, an AI-assisted toolset for automating the migration of legacy monoliths to modern, cloud-native microservices.

Chief Technology Officer

Authentidate Sports, New York, NY

2000 - 2002

- Engineered a first-to-market streaming media platform featuring advanced encryption and high-availability failover.
- Directed technical strategy for investor relations and contributed to foundational patent filings for secure media delivery.

Principal Consultant - Architect

Cap Gemini Ernst & Young, New York, NY

1999 - 2000

- Architected mission-critical trading systems and e-commerce platforms for Fortune 500 clients, including **Reuters, Deutsche Bank, and Chase**.
- Appointed to the elite SBA1 Architecture Council, advising on global architectural standards.

Chief Designer, OPC Tracker Agent

IBM, Global

1992 - 1999

- Architected Agent-Oriented distributed applications for enterprise-scale systems management across heterogeneous environments.

Open Source & AI Projects

170+ public repositories at github.com/xbill9. *Selected AI-focused work:*

- **AI Education: way-back-home** — an immersive AI workshop platform teaching Google Cloud AI agent development (Gemini, Veo, ADK, MCP, A2A) through a narrative rescue mission, used in Build with AI keynotes and workshops.

- **Google Codelab Sample & Starter Kits: currency-agent** family (Python and Rust) — A2A + ADK + MCP sample agent **featured in a Google codelab**, with AG-UI frontend variants; **adk-hello-world** ADK starter kits in Python, Go, and Java; **a2a-inspector** — web-based tool for debugging and validating A2A protocol servers.
- **Multi-Agent Systems (A2A + ADK): multi-agent** — AI Course Creator, a distributed multi-agent system of specialized microservice agents (research, judge, build) orchestrated over the A2A protocol; **a2a-benchmark** and **a2a-multicloud** — cross-language (Python, Go, TypeScript, Rust) and cross-cloud (GCP, AWS, Azure) A2A performance benchmarking suites.
- **Published Models — First Community Port of Gemma 4 to AWS Inferentia2:** Compiled and published the first community Gemma 4 models for AWS Inferentia2 (Neuron) on Hugging Face — **gemma-4-12B-it-inferentia2**, **gemma-4-E4B-it-inferentia2**, and **gemma-4-E2B-it-inferentia2** (huggingface.co/xbill9) — with companion Docker images at **~1,000 combined pulls** (hub.docker.com/u/xbill9) and a published field report.
- **TPU & LLM Serving: TPU Sprint MCP Toolkit** (aisprintapr2026) — nine specialized MCP servers that give LLMs "eyes and hands" for deploying and optimizing workloads on Cloud TPU v6e and serverless GPUs (Cloud Run L4), companion to published Gemma-on-TPU deployment guides; **gemma4-tips** — self-hosted Gemma 4-powered DevOps/SRE agents served via vLLM on Cloud Run GPUs, with AWS and Azure variants.
- **Agentic Coding Tool Integrations:** MCP servers and agents built for and ported across **Claude Code, OpenAI Codex, Gemini CLI, Antigravity CLI, and Kiro** — Looker BI (**looker-mcp-claude**), Gemini image and video generation (**nb2lite**, **omni-flash-video-agent**), Google Workspace (**workspace-mcp**), and Google Developer Knowledge (**knowledge-api**); plus production-grade Rust MCP servers (stdio and streaming-HTTP on Cloud Run with Identity-Aware Proxy) and "Cymbal Superstore" MCP implementations across ~19 programming languages.

Speaking & Community Leadership

- **Speaker | GDG Queens NY DevFest - AI on Edge | Oct 2026:** Featured talk, **"Gemma 4 Tips and Tricks"**.
- **Speaker | LLM Day NYC | June 2026:** Live demo and talk on **Gemma 4**.
- **Keynote Speaker | Hack Michigan - Way Back Home | May 2026:** Keynote on multimodal, real-time AI agents.
- **Keynote Speaker | Build with AI - Way Back Home | March 2026:** Keynote on AI agent development and the evolution of the software agent paradigm.
- **Speaker | UofT Build with AI, GDGoC UofT Scarborough | Mar 2026:** Presented **Multimodal ADK with Live Models**.
- **Speaker | Agent Glow Hackathon | Mar 2026:** Presented **Multimodal Gemini Live** and Vertex AI integration.
- **Speaker | Detroit Black History Month Innovation Summit | Feb 2026:** Presented **"LLM Deployment"** on enterprise MLOps and Gemini/Vertex AI integration.
- **Additional Workshops & Mentoring:** Build with AI Workshops (GDG Queens 2026, DevFest Brooklyn/Queens 2025), Future of Data and AI: Agentic AI Conference (2026), Self Improving Agents Hack NYC (2026); Technical Mentor for Agentverse NYC, Accelerate AI with Cloud Run, and Build with AI NYC (2025).

Education

Master of Science (M.Sc.), Computer Science

University of Western Ontario

- *Thesis: Distributed Application Management and Extensible Agent Technology*
- *Recipient of the NSERC Canadian Post Graduate Scholarship.*

Honours Bachelor of Science (B.Sc.), Computer Science

University of Western Ontario

Publications

300+ technical articles published across dev.to and Medium. Highlights:

- **"Cross Language A2A Agent Development and Testing"** - Google Cloud Blog
- **"Building AI Agents with the GO Agent Development Kit (ADK)"** - Google Medium
- "**TPU Deployments with Gemma 31B, v6e-8, and Antigravity CLI**" - Medium
- "**Porting Gemma-4 (2B/4B/12B) to AWS Inferentia2**" - dev.to
- "**MCP Configuration for Looker with Claude Code**" - dev.to
- "**Gemma-SRE: Self-Hosted vLLM Infrastructure Agent**" - Medium
- "**Extending a MCP/A2A Currency Agent with AG-UI and Antigravity CLI**" - Medium
- "**Debugging Deployments with Gemma 12B, TPU v6e-1, MCP, and Antigravity CLI**" - Medium
- "**Cross Cloud A2A Agent Benchmarking with Antigravity CLI**" - Medium